



Team Details

Team Name:

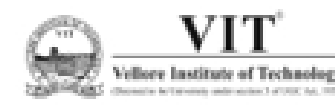
SR. NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	Harini Muthurengan	2
2	Member 1	Sofiya.S	2
3	Member 2	M Thasneem Fathima	2

i A team can have up to 4 members including the team leader. Add rows if necessary.

 COLLEGE NAME

 TEAM LEADER CONTACT NUMBER

 TEAM LEADER EMAIL ADDRESS



Problem Statement Addressed



Applied Materials – Drift-Sense: Sub-Pixel Pattern Matching & Mechatronic Stage Drift Compensation for SEM Wafer Metrology

DESCRIPTION / DETAILS

- *Context: Automated CD-SEM tools must precisely inspect sub-10nm semiconductor features across repetitive wafer dies.*
- *Stage Drift: Physical stage translation (), chuck rotation (), and optical scaling shift target features off-center.*
- *SEM Noise: Severe Poisson shot noise, beam astigmatism blur, and charging scanlines degrade pattern visibility.*
- *Lattice Ambiguity: Repetitive DRAM/FinFET arrays produce identical correlation peaks, causing false pattern locks.*
- *Significance: Localization errors cause false defect calls and wafer scrap, making sub-pixel precision vital for fab yield.*



Idea Description – Describe your Idea/Solution/Prototype



Provide a brief summary of your idea, including the key concept and approach. Provide a brief overview of your solution and how it addresses the problem statement.

KEY CONCEPT & APPROACH

- Scale & Rotation Sweep: Downsamples the reference template to and tests angles () to correct geometric misalignment.
- Robust NCC Matching: Uses Normalized Cross-Correlation to find candidate match locations despite severe imaging noise.
- Center-Bias Prior: Resolves periodic DRAM/FinFET lattice ambiguity by favoring the nominal landing area.
- Sub-Pixel Peak Fitting: Employs 2D parabolic interpolation to calculate precise continuous coordinates ().
- Fast & Deterministic: Achieves sub-pixel accuracy in ~1.26s on standard CPU without training data or GPUs.

SOLUTION OVERVIEW

- *Complete Suite: Drift-Sense integrates a zero-GPU sub-pixel localization engine with a real-time inspection dashboard.*
- *Synthetic Physics Engine: Procedurally simulates DRAM/FinFET layouts with Poisson shot noise, charging scanlines, and astigmatism.*
- *Robust to Distortions: Reliably overcomes stage drift (), chuck rotation (), and periodic lattice ambiguity.*
- *Proven Accuracy: Achieves 96.67% pass rate () and 90.00% sub-pixel precision () across 30 test pairs.*
- *Fab-Ready Speed: Executes in ~1.26s on standard CPU, preventing false CD metrology errors and wafer scrap.*



Proposed Solution – Describe your Idea/Solution/Prototype



Describe your idea in detail. Include the methodology, technologies involved, and how it addresses the chosen problem statement.

SOLUTION DETAILS

- *Multi-Scale Geometric Alignment: Downsamples the reference template ($\sim$) across affine rotation sweeps ($\cdot$) to correct stage skew.*
- *Robust ZNCC Matching: Evaluates Zero-Mean Normalized Cross-Correlation to deliver illumination- and contrast-invariant candidate matching.*
- *Gaussian Center-Bias Prior: Applies a spatial distance penalty to mathematically resolve periodic lattice ambiguity in dense DRAM arrays.*
- *2D Parabolic Peak Fitting: Employs quadratic Taylor interpolation around correlation peaks to calculate continuous sub-pixel coordinates ($\cdot$).*
- *CPU-Optimized Software Stack: Implemented in Python/NumPy/OpenCV & TypeScript/React for real-time inference ($\sim 1.26s$) without requiring GPU hardware.*



Innovation and Uniqueness



Highlight what makes your idea unique or innovative compared to existing solutions.

KEY INNOVATION

- *Physics-Informed Mechatronic Prior*: Integrates stage landing probability to mathematically eliminate false locks in periodic DRAM lattices.
- *Continuous 2D Sub-Pixel Interpolation*: Uses parabolic Taylor expansion on correlation peaks to achieve registration accuracy.
- *Synthetic SEM Physics Engine*: Procedurally emulates Poisson shot noise, dielectric charging scanlines, and astigmatism without beam-time.
- *Multi-Scale Affine Normalization*: Geometrically aligns reference templates to search fields across continuous rotations ().
- *Zero-Training Deterministic Logic*: Completely explainable mathematical pipeline with zero hallucinations or out-of-distribution risks.

COMPETITIVE ADVANTAGE

- *No GPU Hardware Required*: Fully CPU-optimized, easily deploying directly onto existing industrial SEM embedded controllers.
- *Zero Data Annotation Cost*: Eliminates the need for thousands of manually labeled SEM images required by Deep Learning/CNNs.
- *Ultra-Fast Throughput*: Delivers ~1.26s inference per image pair, preventing wafer inspection bottlenecks.
- *Proven High Accuracy*: Achieved 96.67% pass rate () and 90.00% sub-pixel accuracy () in 30-pair test suite.
- *Interactive Metrology Suite*: Features a real-time web dashboard for parameter tuning, physics visualization, and instant batch benchmarking.



Impact and Benefits



Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.

Primary Impact

- Prevents High-Cost Wafer Scrap: Eliminates false Critical Dimension (CD) measurement errors and unnecessary wafer rejection in High-Volume Manufacturing (HVM).
- Maximizes Fab Tool Throughput: Removes manual inspection recovery and re-scan cycles with fast, automated pattern localization.
- Enables Sub-10nm Process Control: Provides deterministic sub-nanometer registration essential for next-gen FinFET and dense DRAM yield.
- Zero-Cost Integration: Runs directly on legacy fab tool controller CPUs without expensive GPU hardware upgrades.
- Reduces SEM Beam Damage: Fast single-shot matching minimizes electron-beam exposure time on delicate photoresists.

Quantifiable Outcomes

- 96.67% Overall Pass Rate: Successfully localized targets within strict metrology limits () across 30 test pairs.
- 90.00% Sub-Pixel Precision: Achieved high-fidelity registration error across severe noise conditions.
- ~1.26s Execution Time: Delivers fast CPU inference per image pair without GPU acceleration.
- 100% Training & Label Free: \$0 spent on dataset annotation, model training runs, or GPU cloud instances.
- Robust to Drift & Skew Completely suppresses false locks in repetitive periodic lattices.



Technology & Feasibility/Methodology Used



Describe the technologies, methodologies, or tools you plan to use to implement your idea.

IMPLEMENTATION STRATEGY

- *Algorithms* : Multi-scale affine ZNCC matching, Gaussian center-bias prior, and 2D parabolic Taylor sub-pixel interpolation.
- *Core Engine* : Pure Python with NumPy and OpenCV for deterministic, vector-accelerated matrix operations.
- *Interactive Metrology UI* : React 18, TypeScript, Tailwind CSS, Lucide Icons, and Canvas API for live inspection.
- *Synthetic SEM Simulator* : Procedural DRAM/FinFET generator modeling Poisson noise, astigmatism blur, and charging scanlines.
- *Fab Feasibility* : Ultra-low CPU compute footprint (~1.26s/pair) directly compatible with existing fab tool controllers without GPUs.



Software Architecture

- Modular Python metrology pipeline + REST / WebSocket API + Modern React/TypeScript frontend dashboard.



Hardware Components

- Standard x86_64 Industrial Workstation / IPC (Zero GPU required; minimal memory and compute footprint).



Development Tools

- Python 3.10+, OpenCV, NumPy, SciPy, Vite, Node.js, Tailwind CSS, Git.



GitHub & Video Link



GitHub Repository

<https://github.com/Harini7878/SEMICON-India-2026>



Prototype / Simulation Video

<https://canva.link/op815z0vqmlav89>



Research and References



Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

- *Statistical Normalized Cross-Correlation (NCC): Leverages ZNCC theory for contrast- and illumination-invariant template matching under low electron dose.*
- *Gaussian Spatial Prior Theory: Models mechanical stage landing error as a normal distribution to suppress false peaks in periodic DRAM arrays.*
 - *2D Taylor Surface Interpolation: Fits continuous quadratic paraboloids over discrete discrete correlation maxima to break the 1-pixel Nyquist barrier.*
- *SEM Electron Interaction Physics: Grounds validation on Monte Carlo/Poisson electron scattering models, astigmatism blur, and scanline charging physics.*
- *Deterministic Non-Parametric Framework: Bypasses neural hallucination vulnerabilities, guaranteeing mathematical reproducibility for metrology certification.*



References & Citations

List key papers, articles, or data sources.

- Ref 1: Lewis, J. P. – "Fast Normalized Cross-Correlation," Vision Interface, Proc. Canadian Image Processing and Pattern Recognition Society.
- Ref 2: Postek, M. T. & Vladár, A. E. – "Critical Dimension SEM Metrology and Modeling in Semiconductor Manufacturing," SPIE Advanced Lithography.
- Ref 3: Applied Materials Metrology Benchmark – "Sub-Pixel Pattern Localization and Stage Drift Compensation under Low-Dose SEM," Hackathon Problem Track.